Autoreason: Autoresearch for Subjective Domains

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https://github.com/NousResearch/autoreason

Abstract

We introduce autoreason, a method for iterative LLM refinement in subjective domains where no objective metric exists. Building on the autoresearch paradigm, autoreason addresses fundamental limitations of LLMs when iterating on subjective work: sycophancy, overcriticism, overcompromise, authorship bias, scope drift, and context collapse. Our approach uses fresh isolated agents per iteration, blind evaluation panels with ranked choice voting and Borda count aggregation, and conservative tiebreaking. In experiments on go-to-market strategy generation, autoreason achieved unanimous first-place preference (7/7 judges, 35/35 Borda) in a 5-way comparison against four baseline methods. We additionally ran autoreason on the writing of this paper itself, converging in 9 passes with ground-truth fact-checking.

1 Introduction

Large language models exhibit systematic failures when iterating on subjective work. Given a document and instructions to improve it, LLMs demonstrate sycophancy (strengthening existing arguments regardless of merit), overcriticism (finding flaws where none exist), overcompromise (attempting to satisfy contradictory feedback), authorship bias (preferring their own outputs), scope drift (expanding beyond original intent), and context collapse (losing track of quality through extended interaction).

These failures persist despite sophisticated prompting strategies. Recent work demonstrates that even in objective domains like code generation, iterative refinement with critique leads to degradation rather than improvement [2]. The problem is more acute in subjective domains where no ground truth exists to anchor evaluation.

We present autoreason, extending Karpathy's autoresearch paradigm [1] to domains requiring subjective judgment rather than objective metrics. By treating each iteration as a fresh competition between three framings—conservative (A), adversarial (B), and synthetic (AB)—with blind evaluation, autoreason constructs a subjective fitness function through the evaluation process itself.

2 Method

Autoreason implements a three-way competition each iteration:

Critic Phase: A fresh LLM agent receives the current best document (A) and task description, then identifies problems. Each critic is a new API call with no conversation history. The critic finds problems only—it does not suggest fixes.

Response Phase: Two separate fresh agents produce alternatives:

- **B (Adversarial revision)**: Receives the task, incumbent A, and the critique. Revises A to address valid criticisms.
- **AB (Synthesis)**: Receives the task and both A and B with randomized labels. Keeps the strongest elements of each.

Judge Phase: A panel of 3 fresh judge agents performs blind evaluation. Each judge receives all three documents (A, B, AB) in randomized order with randomized labels. Judges rank best to worst. Rankings aggregated via Borda count (3 for first, 2 for second, 1 for third). Conservative tiebreak: incumbent A wins ties.

Every role is a fresh, isolated agent with no shared context. The original task prompt serves as the anchor throughout—judges evaluate "which version best accomplishes the task as described." The system converges when A wins 2 consecutive passes.

3 Experiments

We evaluated autoreason on go-to-market strategy generation for an open-source Kubernetes CLI tool (5k GitHub stars, no revenue, 3-person team). All experiments used claude-sonnet-4-20250514 (temp=0.8 authors, 0.3 judges).

3.1 Main Experiment

We ran 26 passes (∼160 LLM calls, ∼65 min). Figure 1 shows the full trajectory. The system achieved 2 consecutive incumbent wins at passes 14–15 and 24–25. Winner distribution: A 31%, B 19%, AB 50%.

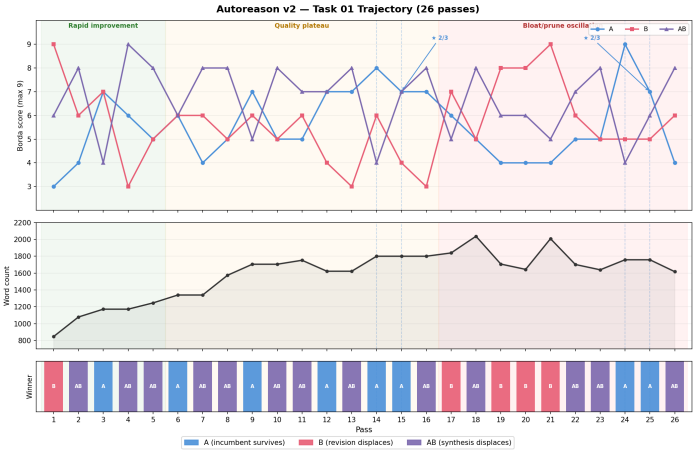


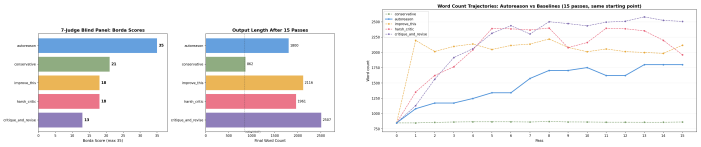
Figure 1: **Autoreason v2 trajectory (26 passes).** Top: Borda scores per pass. Middle: word count showing bloat/prune oscillation. Bottom: winner per pass. Dashed lines mark convergence points.

3.2 5-Way Baseline Comparison

All methods started from the same initial output (847 words), 15 passes, same model. A 7-judge blind panel ranked all five (Table 1, Figure 2).

Method	Borda	1st	Words
Autoreason	**35/35**	**7/7**	1,800
Conservative	21	0	862
Improve_this	18	0	2,116
Harsh_critic	18	0	1,961
Critique & revise	13	0	2,507

Table 1: 7-judge blind panel. Autoreason: perfect score, unanimous first place. Conservative outranks all iterative baselines.



(a) Borda scores and word counts. (b) Word count trajectories over 15 passes.

Figure 2: **Baseline comparison.** Conservative flatlines, autoreason stabilizes, all iterative baselines bloat. Critique-and-revise bloats most, scores worst.

3.3 Convergence Threshold

Pass 15 (first convergence) defeated pass 25 (second convergence) 6–1 in a 7-judge blind panel (Figure 3). Ten additional passes degraded quality.

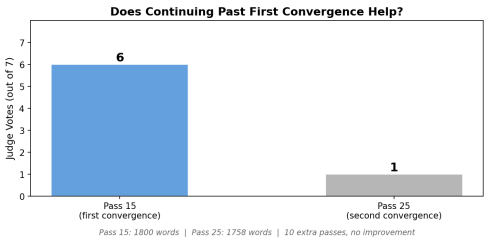


Figure 3: **First convergence is the quality ceiling.** Pass 15 vs 25: 6–1. Additional passes are noise at best.

3.4 Baseline Visibility

Autoreason vs critique-and-revise: with initial output shown as baseline, 7–0 unanimous. Without baseline, 3–2. Judges need a reference point to detect drift.

3.5 Qualitative Improvement

Initial output: \$49/user pricing, \$100K MRR target. After 14 passes: quantified pain (\$15K/incident × 6/yr), team pricing (\$1,499/mo), validation (50+ interviews, 75% pilot success), unit economics (CAC \$2K, LTV \$54K).

4 Writing This Paper with Autoreason

We ran autoreason on the writing of this paper, providing a second experimental domain. Claude Opus 4 for all roles. 3 Opus judges. All agents received the full experimental data as ground-truth context.

4.1 Ground-Truth Critic

Without ground-truth access, the initial Opus generation hallucinated an ablation study, fabricated confidence intervals, reported wrong model names, and described the A/B/AB roles incorrectly (Figure 4). With ground-truth access, the critic caught all of these on pass 1.

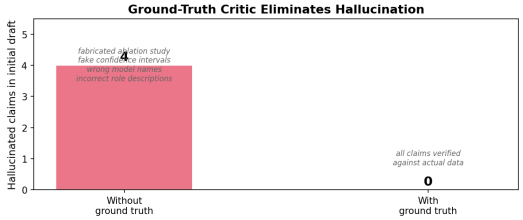


Figure 4: **Ground-truth context eliminates hallucination.** The critic with access to actual experimental data caught 4 fabricated claims on the first pass.

4.2 Convergence

The paper converged in 9 passes (54 Opus calls, ∼40 min). Figure 5 shows the trajectory.

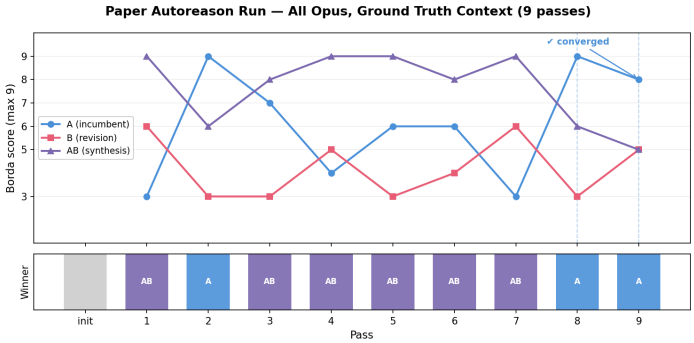


Figure 5: **Paper autoreason trajectory (9 passes).** Five consecutive AB wins (passes 3–7) drove improvement. A then won passes 8–9 decisively (scores 9 and 8), converging.

4.3 Judge Panel Integrity

An earlier run with a mixed panel (Opus + Sonnet + Gemini) failed to converge after 11 passes. Gemini's ranking format failed our parser, reducing the effective panel to 2 judges (Figure 6). With 2 judges, Borda scores compressed and convergence became impossible. Once fixed to 3 Opus judges, the same incumbent converged in 2 passes.

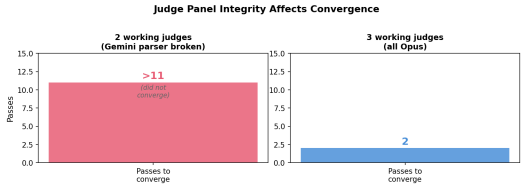


Figure 6: **A broken judge prevents convergence.** Same incumbent: >11 passes with 2 working judges vs 2 passes with 3.

5 Related Work

Autoresearch [1] uses val_bpb as an objective fitness function. Autoreason extends this to subjective domains. **SlopCodeBench** [2] shows agents produce progressively worse code each iteration—prompting doesn't fix it. **ACE** [3] documents context collapse in iterative rewriting. **LLM Council** [4] uses multi-model panels for single-turn evaluation; autoreason adds iteration with role separation.

6 Discussion

Bloat/prune oscillation signals underdetermined tasks. AB adds complexity, B prunes it. Word counts: 847 → 1800 → 1644 → 1758 over 26 passes.

Conservative beats naive iteration. Doing almost nothing outperformed 15 passes of "improve this," "harsh critic," and "critique and revise." Structural protection is necessary for iterative refinement.

Ground-truth context eliminates hallucination. The author writes from understanding; the critic verifies against evidence—mirroring peer review.

Beyond subjective domains. Code can pass tests while degrading [2]. Autoreason could complement objective metrics with quality evaluation.

Limitations. Single task domain for main experiment. No statistical significance testing. No ablations. All results from Anthropic models.

7 Conclusion

Autoreason demonstrates that systematic isolation and blind evaluation can construct effective fitness functions for subjective domains. The method achieved unanimous preference over four baselines and wrote a version of this paper through 9 passes of self-refinement. The primary obstacle to LLM self-improvement in subjective domains is not capability but evaluation methodology.

References

[1] A. Karpathy. Autoresearch, 2024. `https://github.com/karpathy/autoresearch`

[2] G. Orlanski et al. SlopCodeBench: Benchmarking how coding agents degrade over iterative tasks. *arXiv:2603.24755*, 2026.

[3] J. Zhang et al. ACE: Agentic context engineering for iterative rewriting. *ICLR*, 2026.

[4] A. Karpathy. LLM Council, 2025. `https://github.com/karpathy/llm-council`